

FungMod: provenance-aware virtual experiments with identifiability analysis, preregistered model criticism and cross-solver reproduction for fungal substrate degradation

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Draft status (2026-10-06). Software and methods paper draft assembled from the recorded studies in the repository. Every number below is taken from a checksummed result file named in the text; the tables are generated L^AT_EX fragments and the figures generated PDF files, both written by one command from those files. Items marked *pending* are not yet recorded. Human review of the biological interpretation is outstanding and is not asserted here. This draft is not a submission.

1 Summary

FungMod is a Python package for mechanistic virtual experiments on fungal and enzyme-mediated substrate degradation. A registry of organisms, substrates, enzymes, environments and parameter records with explicit provenance and maturity labels is composed through case templates into well-mixed process models; a compiled numerical core integrates them with build-time unit resolution; calibration, Bayesian sampling with identifiability classes, and preregistered model criticism run on the same models; and the models export to SBML, SED-ML, COMBINE and P_Etab so that independent tools can simulate and fit them. The package is organised around one rule: a simulation may report only what its inputs and their evidence support. Unknown inputs stay unknown, exploratory assumptions are labelled, software verification is never presented as empirical validation, and every scientific study is run under a plan frozen and digested before the first fit.

This paper describes the design and demonstrates it on one real organism: the batch cultures of *Trichoderma harzianum* P49P11 on particulate cellulose published with the Gelain et al. deposit (Gelain et al., 2020). We report (i) a retrospective whole-condition holdout benchmark of the published hydrolysis candidate, (ii) a posterior-sampling study that classifies which of its nine constants the published duplicate means identify, (iii) a preregistered comparison of three added mechanisms, none of which the data support under the frozen rules, and (iv) a cross-solver reproduction in which COPASI (Hoops et al., 2006) simulates the exported P_Etab problem (Schmiester et al., 2021) to within 1.7×10^{-8} of the observation scale and then finds an objective 1.3 percent below FungMod’s recorded optimum, which FungMod evaluates to the same value. The last result is an honest finding against our own optimiser’s stopping rule, and the paper reports it as such.

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2 Statement of need

Models of fungal degradation are usually published as bespoke equation sets fitted to one dataset, with units, parameter provenance and the error model left implicit. Reusing such a model for another strain, substrate or reactor requires re-deriving all three. General systems-biology tools (COPASI, SBML-based simulators, PETab-based estimation frameworks) solve the simulation and estimation problems well but do not carry the biological bookkeeping that a degradation study needs: which constant came from which assay under which conditions, which assay units cannot be converted to molarity without a specific-activity measurement, which output is a measured pool and which is a closure ledger, and which claim the evidence actually licenses. FungMod supplies that bookkeeping as code, keeps it attached to every output, and hands the resulting problem to the general tools through the community standards rather than replacing them.

3 Design

3.1 Registry, case templates and maturity

Every biological input is a registry record with a source, a confidence level, a validity range and notes. Case templates compose records into a model config for a (fungus, substrate, environment) triple: state roles, initial-state mappings, product maps with stoichiometric coefficients that may be bound to parameter records (a biomass yield Y and its complement $1 - Y$), and a list of generic process templates. The first organism case composes six generic processes (enzyme-explicit Michaelis-Menten consumption with a yield split, three first-order losses and two producer-proportional, saturably induced activity syntheses) with no organism-specific code path. Assay activities (filter-paper units and beta-glucosidase units) are independent base dimensions of the unit registry: they cannot be converted to mass or molarity, and a conversion would have to enter as a sourced parameter.

A run carries a mode (`scientific` or `exploratory`) and every record a maturity. The scientific mode refuses inputs below its maturity floor; the exploratory mode runs them and labels the output. The result bundle lists mechanisms, assumptions, provenance, limitations and suggested experiments next to the trajectories.

3.2 Compiled process core

Processes declare required and changed states with units and a rate law on unit-bearing quantities. The compiled core probes the stoichiometric matrix from each process’s contributions, checks linearity in the rate, resolves every unit conversion once at build time, and integrates $dy/dt = N v(t, y)$ on plain floats with SciPy’s LSODA (Virtanen et al., 2020). Every shipped process and modifier supplies a numeric kernel that reproduces the unit-aware arithmetic in the same operation order; a process without one uses the recorded unit-aware fallback, never silently. Trajectories and evaluation counts are identical to the unit-aware path on every packaged configuration, and solves are 6 to 60 times faster, which is what makes posterior sampling and multi-start holdout studies affordable.

3.3 Calibration, posterior sampling and identifiability classes

A configured-condition predictor rebuilds the model config for each candidate parameter vector, so a yield bound to a fitted parameter reaches every place the public path puts it. On top of it: multi-start least squares in log-parameter space with whole-condition holdouts and a practical-rank diagnostic; an affine-invariant ensemble sampler (Goodman and Weare, 2010) with checkpointing, autocorrelation-based convergence rules and a shared observation-noise

multiplier sampled jointly with the parameters; identifiability classes derived from the posterior credible interval against the prior box (`identified`, `weakly_identified`, `bounded_above_only`, `bounded_below_only`, `prior_dominated`, with declared thresholds); local Fisher information at the best sample; and posterior predictive bands and coverage. The classes are the operational answer to “what do these data identify” that profile-likelihood analysis (Raue et al., 2009) gives for a single optimum.

3.4 Frozen plans and the claim boundary

Each scientific study is specified by a JSON plan with the data digests, models, bounds, error model, decision rules, outcome vocabulary and excluded claims. The plan’s SHA-256 is pinned by a test; a change requires a dated amendment inside the file before any affected run, and every result cites the digest it ran under. The evaluator that could authorise a publication claim stays false until replicate-level or independent data exist.

3.5 Standards and cross-solver reproduction

Models export to SBML Level 3 Version 2 (Hucka et al., 2003) with explicit numeric unit conversions in the kinetic laws, to SED-ML (Waltemath et al., 2011) and COMBINE archives (Bergmann et al., 2014), and to P_Etab (Schmiester et al., 2021). Two representation choices keep the exported model faithful where SBML has no construct: a product coefficient bound to a parameter is written as a separate reaction whose kinetic law multiplies the process rate by the parameter (or its complement), so the dependence stays live; and assay-activity units are written as named dimensionless unit definitions listed in the model notes, with no SI equivalent implied. A reference simulator independent of the production path checks every export, and a COPASI driver imports the P_Etab problem, corrects the column weights that COPASI’s importer stores as σ rather than $1/\sigma^2$, simulates, fits, and re-evaluates every optimum on the P_Etab objective from COPASI’s own time courses.

4 Results on the Gelain 2020 cultures

All results are retrospective analyses of published duplicate means that already informed model development. None is blind, independent or a validation of biology. Sources: the directories `gelain_2020_v2`, `gelain_2020_bayesian`, `gelain_2020_criticism` and `gelain_2020_petab` under `data/benchmarks` in the repository, each with a plan, a README and digested result files.

4.1 Whole-condition holdouts of the published candidate

Table 1 lists every model, family and scenario of the joint benchmark with its held-out error and screen verdict, and Figure 1 shows the frozen held-out predictions of the hydrolysis candidate and of the re-fitted published equations against the published means on the three cellulose loadings. The joint benchmark compares seven model and family combinations against 144 published means (biomass, cellulose or glycerol, filter-paper activity and beta-glucosidase activity) with each condition held out in turn. On the three cellulose loadings the activity-driven hydrolysis candidate reaches a mean normalised held-out mean squared error of 0.092 (training-maximum normalisation) against 0.0062 for the published equations re-fitted with their own structure; the published structure fails the complexity screen that the plan declares, the hydrolysis candidate passes it under the correlated error assumption only. 33 of 33 folds pass the numerical checks; 6 of 132 optimiser starts fail and are reported. No model is promoted to validated.

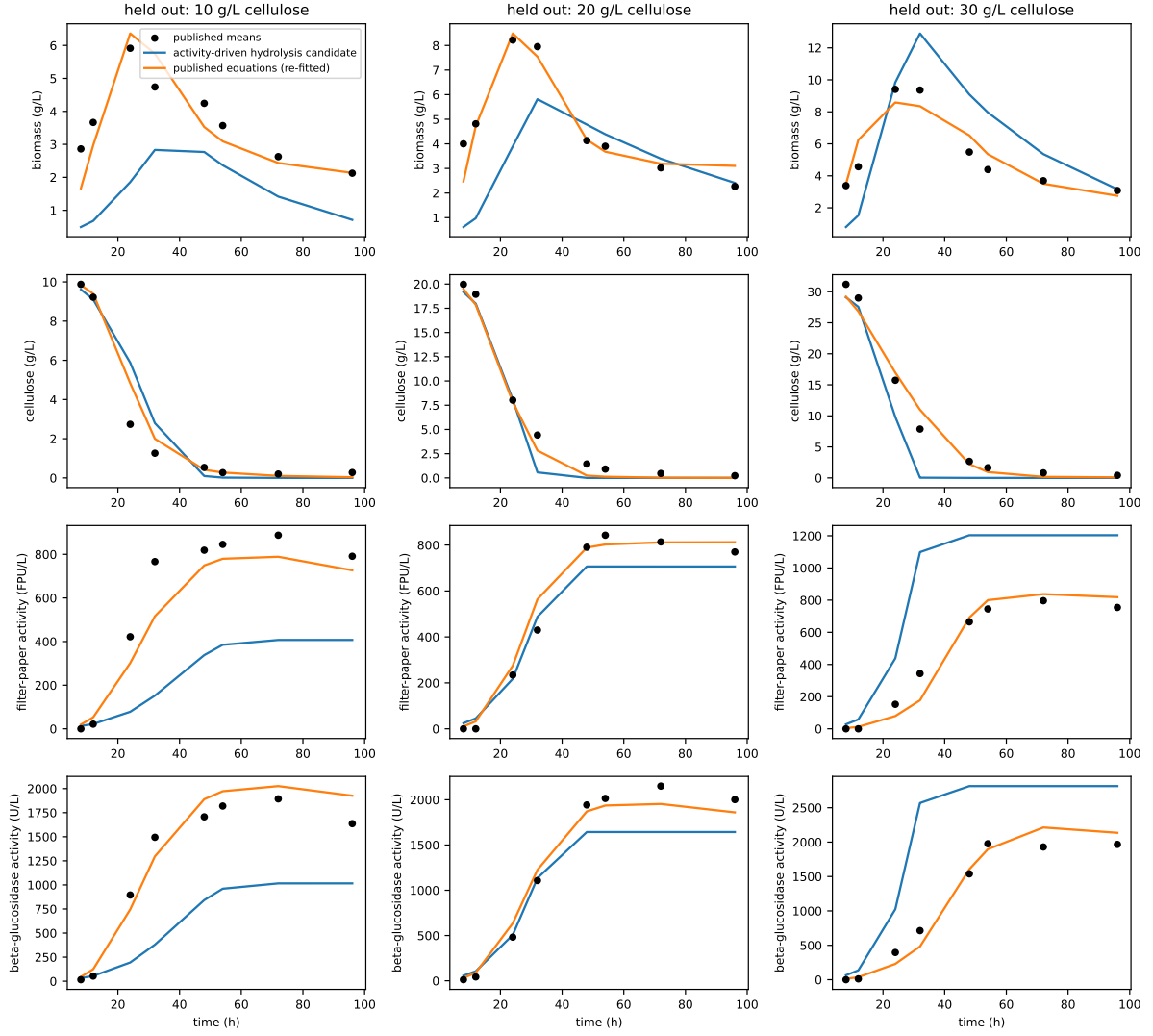


Figure 1: Whole-condition holdouts of the joint benchmark on the three cellulose loadings. Published means (points) against the frozen held-out predictions (lines) of the activity-driven hydrolysis candidate and the re-fitted published equations, each loading predicted from a fit to the other two (primary scenario). Measurement error is unknown; the points are published means.

Table 1: Whole-condition holdouts of the joint Gelain 2020 benchmark (v2)

Family	Model	Scenario	Parameters	Practical rank	Mean normalized held-out MSE	Complexity screen
glycerol	effective	primary	4	4	0.03287	passed
glycerol	published	primary	5	5	0.00594	passed
glycerol	retained	primary	6	6	0.02925	passed
cellulose	effective	primary	8	8	0.1095	passed
cellulose	published	primary	20	19	0.006185	failed
cellulose	hydrolysis	primary	9	9	0.09183	failed
cellulose	hydrolysis_ retained	primary	11	11	0.09316	failed
cellulose	effective	correlated_ assumption	8	8	0.1139	passed
cellulose	published	correlated_ assumption	20	18	0.007541	failed
cellulose	hydrolysis	correlated_ assumption	9	9	0.09497	passed
cellulose	hydrolysis_ retained	correlated_ assumption	11	11	0.09763	passed

Folds that passed the numerical checks: 33 of 33; optimiser starts that failed: 6 of 132. Validated: False (retrospective comparison on published means).

4.2 What the data identify

Table 2 gives the class, posterior median, credible interval, autocorrelation time and effective sample size of every constant; Figure 2 shows the posterior median and the 5 to 95 percent band of the prediction on each loading, with the published means. Posterior sampling of the nine registry constants of the hydrolysis candidate (24 walkers, 24 000 steps, 4 000 burn-in, log-uniform priors on the declared bounds, one shared noise multiplier) converges by the declared rule (every autocorrelation estimate reliable, effective sample sizes 1 414 to 2 119, mean acceptance 0.325). The data identify five constants (the consumption capacity k_h , the yield Y , the biomass loss rate kd and both specific production rates qF , qB , with 95 percent credible intervals of 5 to 16 percent of the prior width in $\log_{10}$), bound the half-saturation constant K_h from below only, and bound the induction constant K_{ind} and both activity loss rates kF , kB from above only. The shared noise multiplier’s posterior median is 2.25 (credible interval 1.95 to 2.63) at an assumed 10 percent error, which says that the candidate cannot fit biomass and cellulose together at that error level. The registry records keep their frozen point values; the study records which must be read as ranges.

4.3 Preregistered model criticism

Tables 3 and 4 carry the holdout screen and the posterior verdicts per mechanism; Figure 3 shows each mechanism’s change in pooled held-out error relative to the baseline against the screen’s thresholds. A frozen plan declared three mechanisms that could reduce the biomass-cellulose misfit, each added to the baseline with its bounds, an error model, whole-condition holdouts with a complexity screen (at least 10 percent pooled held-out improvement, no observable more than 10 percent worse, full practical rank), decision rules and an outcome vocabulary. Stage A (least squares, five starts per fold) was run twice: once with the optimiser as first written, and again under a dated amendment that declared the finite-difference step after the cross-solver check below found the first run stopping above the minimum. The verdicts quoted here are from the second run; the first is kept in the plan’s amendment log and in the ledger. An induction-state memory (M1) leaves the pooled held-out error unchanged (0.1 percent better) and makes cellulose three times worse, not supported; a soluble product pool with Monod uptake and product inhibition (M2) lowers the pooled held-out error by 23 percent in the primary scenario and 26

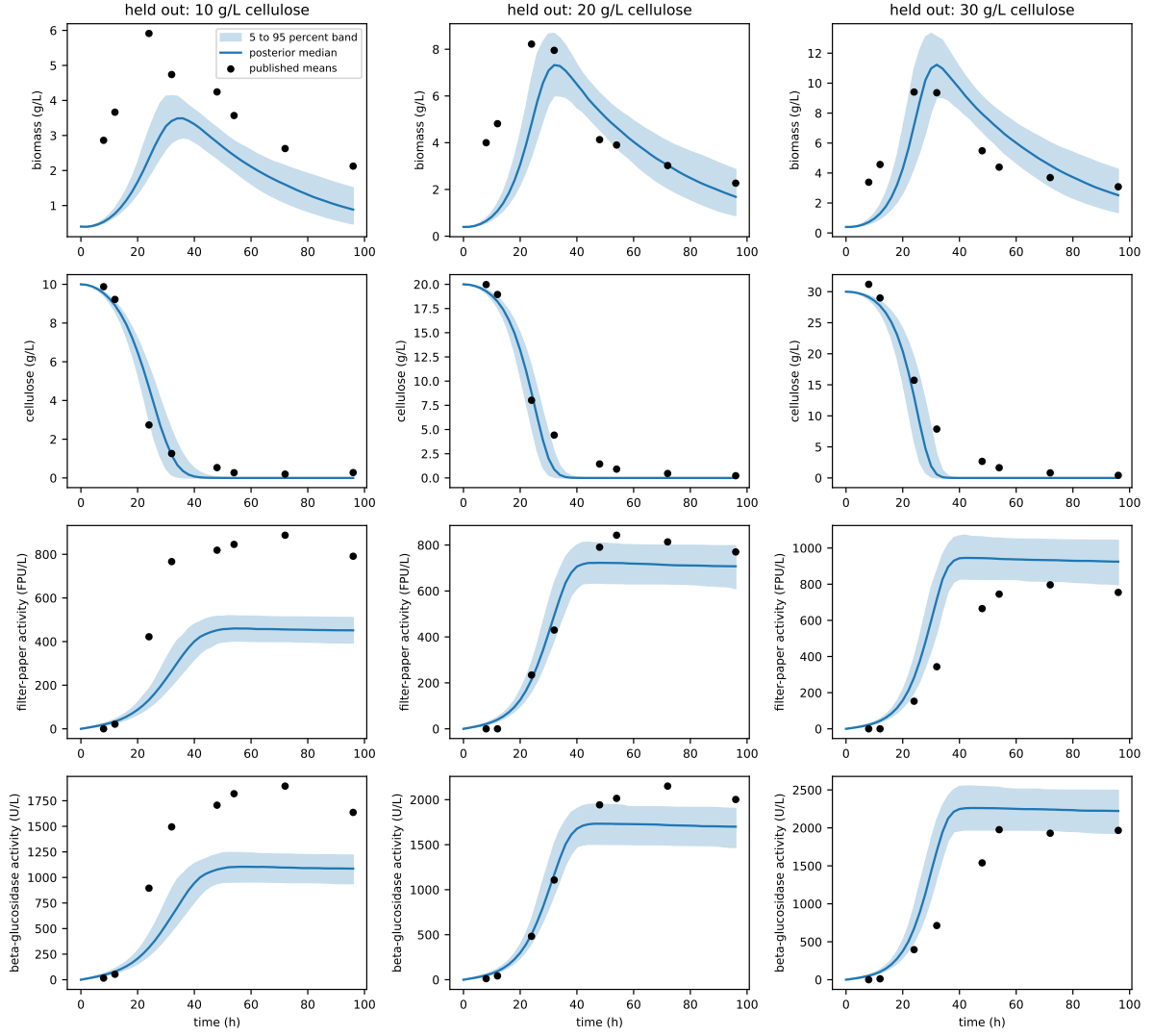


Figure 2: Posterior predictive bands of the hydrolysis candidate on the cellulose loadings. Posterior median and the 5 to 95 percent band of the model prediction over the posterior samples of the nine registry constants (all three loadings fitted; the bands exclude measurement noise), with the published means as points.

Table 2: Identifiability of the nine registry constants of the hydrolysis candidate (BAYES-001)

Constant	Class	Posterior median	95% interval	Prior box	Width / prior width (log10)	tau	ESS
k _h	identified	0.01848	[0.00714, 0.0639]	[1e-06, 1]	0.16	305	1576
Kh	bounded_ below_only	18.37	[4.26, 81.9]	[0.001, 100]	0.26	306	1567
Y	identified	0.4783	[0.328, 0.648]	[0.01, 1]	0.15	299	1606
kd	identified	0.02365	[0.0114, 0.0392]	[1e-05, 0.5]	0.11	274	1754
K _{ind}	bounded_ above_only	0.0188	[0.0103, 0.12]	[0.01, 100]	0.27	289	1659
qF	identified	5.606	[4.2, 7.94]	[0.01, 1e+03]	0.06	324	1483
kF	bounded_ above_only	5.331e-05	[1.21e-06, 0.004]	[1e-06, 0.1]	0.70	278	1724
qB	identified	13.29	[9.9, 18.7]	[0.01, 3e+03]	0.05	339	1414
kB	bounded_ above_only	4.161e-05	[1.21e-06, 0.0028]	[1e-06, 0.1]	0.67	286	1681

Shared noise multiplier: median 2.25, 95% interval [1.95, 2.63] at an assumed 10 percent error. Sampler: 24 walkers, 24000 steps, 4000 burn-in; converged by the declared rule: True; mean acceptance 0.325; effective sample sizes 1414 to 2119.

percent under the correlated error assumption with every observable better (biomass by 25 and 36 percent), passing the screen in both; conversion-dependent accessibility (Kadam et al., 2004) (M3) does not improve the pooled error (0.2 percent worse) with its exponent near its lower bound, not supported. Every fit has full practical rank; every all-condition fit has at least three of the shared constants on their lower bounds, and the M2 fit puts the initial soluble pool on the top of its declared range. In the first run M2 had failed the screen because biomass worsened by 31 percent; that was the stalled optimiser, not the mechanism, and the plan’s outcome for M2 changed from “not supported” to “improves fit but unidentified” when the screen was recomputed. Stage B sampled one all-condition posterior per addition (24 or 28 walkers, 8000 steps, the noise multiplier sampled jointly), each centred on the first-run fit; the chains were not re-run after the amendment because the centre is a starting point, not a result. None of the three chains met the declared convergence rule at that length, so the plan labels their verdicts provisional. Under those labels no addition restores adequacy (every multiplier interval excludes 1.0 and overlaps the baseline’s 2.25); only the induction memory constant of M1 is weakly identified, while the four constants of M2 are prior dominated or bounded on one side and the exponent of M3 is bounded above only, driven towards the baseline; posterior predictive coverage with measurement noise is 91 to 95 of 96 observations. The soluble pool is the one addition the holdouts support, and the one whose constants the duplicate means do not identify; a chain centred on its new fit is the next recorded task.

4.4 Cross-solver reproduction

Table 5 records the gates, the objectives and the parameter agreement. Figure 4 plots the objective at the recorded optimum, after COPASI’s local fit and at the end of its ten random starts. The baseline problem (three loadings, four observables, 96 measurements, nine parameters on a $\log_{10}$ scale with the criticism plan’s bounds, σ equal to each observable’s maximum over the loadings) was exported as a Petab problem and reproduced in COPASI 4.48 under a frozen plan. The first run found a defect in our own optimiser. At FungMod’s then recorded optimum (objective 4.0307) COPASI’s time courses differed from the compiled core by at most 1.7×10^{-8} of σ , but COPASI’s Levenberg-Marquardt fit from the same point reached 3.9803 and the best of ten seeded random starts 3.9768, 1.3 percent lower, and FungMod’s compiled core evaluated that

Table 3: Stage A of the preregistered model-criticism study: whole-condition holdouts and the R1 screen

Model	Scenario	Mean normalized held-out MSE	Change vs M0	Full rank everywhere	Screen (R1)
M0_baseline	primary	0.09184		True	passed
M0_baseline	correlated_ assumption	0.09324		True	passed
M1_induction_state	primary	0.09173	+0.1%	True	failed: pooled normalized held-out error improves by less than the required fraction; an observable worsens by more than the allowed fraction
M1_induction_state	correlated_ assumption	0.09606	-3.0%	True	failed: pooled normalized held-out error improves by less than the required fraction; an observable worsens by more than the allowed fraction
M2_soluble_ product_pool	primary	0.0703	+23.4%	True	passed
M2_soluble_ product_pool	correlated_ assumption	0.0689	+26.1%	True	passed
M3_conversion_ dependent_ accessibility	primary	0.09201	-0.2%	True	failed: pooled normalized held-out error improves by less than the required fraction
M3_conversion_ dependent_ accessibility	correlated_ assumption	0.0979	-5.0%	True	failed: pooled normalized held-out error improves by less than the required fraction; an observable worsens by more than the allowed fraction

Per-observable pooled normalized held-out MSE, primary scenario:

Model	biomass	substrate	cellulase_activity	beta_glucosidase_activity
M0_baseline	0.07863	0.01323	0.1603	0.1152
M1_induction_state	0.06378	0.04101	0.1542	0.1079
M2_soluble_product_pool	0.0592	0.01128	0.127	0.08372
M3_conversion_dependent_accessibility	0.07387	0.01209	0.1648	0.1173

Table 4: Stage B of the model-criticism study: adequacy, identification and coverage per added mechanism

Model	Chain	Converged	R1 holdout	R2 multiplier interval	R3 added-parameter classes	Coverage (95%, with noise)	Outcome
M1__induction__state	24 walkers x 8000 steps	False	False	[1.87, 2.53]	<code>kz_loss</code> weakly identified	91/96	not supported (fails R1) (provisional)
M2__soluble__product__pool	28 walkers x 8000 steps	False	True	[1.65, 2.87]	<code>mu</code> prior dominated, <code>Ks</code> prior dominated, <code>Ki</code> prior dominated, <code>P0</code> bounded_below_only	95/96	improves fit but unidentified (R1, not R3) (provisional)
M3__conversion__dependent__accessibility	24 walkers x 8000 steps	False	False	[1.94, 2.62]	<code>n</code> bounded_above_only	91/96	not supported (fails R1) (provisional)

M0 is the baseline: its all-condition posterior is BAYES-001 (Table 2), which R1 and R3 do not address.

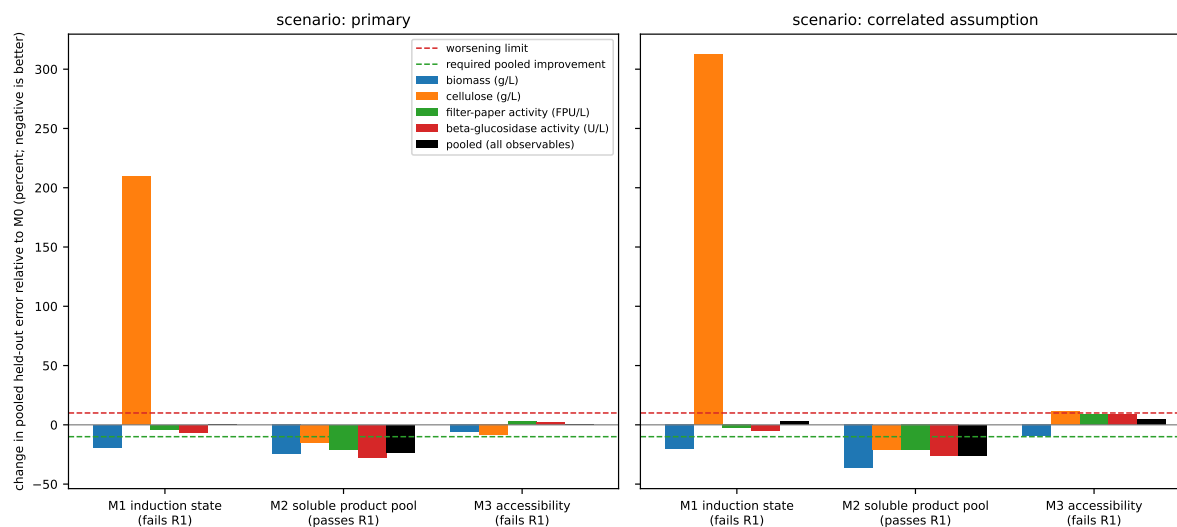


Figure 3: Stage A holdout screen of the model-criticism study. Change in the pooled held-out normalised error of each candidate mechanism relative to the baseline M0, per observable and pooled, in both error scenarios; the dashed lines are the plan’s screen thresholds (at least 10 percent pooled improvement, no observable more than 10 percent worse).

Table 5: Cross-solver reproduction of the baseline optimum in COPASI through PETab (PETAB-001, second run)

Quantity		Value	
Worst abs(COPASI - FungMod) / sigma at FungMod's optimum		1.6e-08	
Objective relative difference at FungMod's optimum		2.1e-08	
FungMod objective (reference)		3.9760719	
COPASI best objective (local_fit)		3.9760718	
Relative difference of the optima		2.5e-08	
Best of the random starts		3.97682 (of 10)	
FungMod at COPASI's best point		3.97607172 (relative difference 1.9e-08)	
Simulation gate / optimum gate		True / True	
Outcome		reproduced	

Parameter	FungMod	COPASI best	Relative difference
k_h	0.01839	0.01839	5e-05
Kh	16.75	16.74	7.4e-05
Y	0.4146	0.4146	1.9e-05
kd	0.01985	0.01985	4.7e-05
K_ind	0.01	0.01	1.1e-14
qF	5.823	5.823	1.3e-06
kF	1e-06	1e-06	6.3e-07
qB	13.94	13.94	4.1e-05
kB	1e-06	1e-06	4.2e-11

point to the same objective (relative difference 7×10^{-9}). The solvers agreed; the optimisers did not. The cause, recorded in a dated amendment of the criticism plan, was the finite-difference Jacobian: with scipy's default step of about 1.5×10^{-8} the differences were dominated by the adaptive ODE integrator's step noise (derivative norms for the weakly entering constants fifteen times their converged values), and the trust region collapsed above the minimum with every start reporting a satisfied step tolerance. Declaring a log-space difference step of 10^{-3} , as the earlier benchmark plan already did, moved every start of the baseline to the same minimum (objective 3.97607, cost spread across five starts 5×10^{-6} , projected gradient norm 2.5×10^{-3} in log space against 0.22 before). Stage A of the criticism study was re-run under the amendment and the cross-solver study was re-run against the new reference: COPASI's local fit now reaches 3.9760718 against FungMod's 3.9760719 (relative difference 2.5×10^{-8}), every parameter agrees to better than 10^{-4} relative, three constants (K_ind, kF, kB) sit on their lower bounds in both solvers, and no random start goes lower. Outcome in the plan's vocabulary: **reproduced**. The episode is reported in full because it is the kind of defect that a cross-solver check exists to find: a result that was internally consistent and wrong by 1.3 percent.

5 Relation to existing tools

COPASI, SBML simulators and PETab-based estimation frameworks are the general infrastructure FungMod exports to, not alternatives to it; the cross-solver study is the demonstration. Genome-scale metabolic reconstructions address intracellular flux, not the extracellular degradation dynamics, assay-defined activities and provenance bookkeeping that FungMod models. Kinetic databases such as SABIO-RK (Wittig et al., 2012) supply enzyme constants; FungMod imports them as sourced records (one pH-response case is bound to a SABIO-RK entry) and keeps their conditions attached.

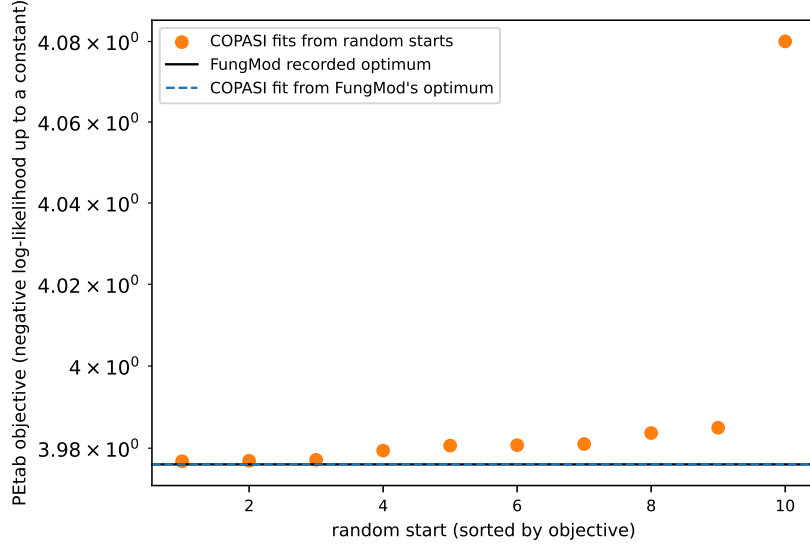


Figure 4: Cross-solver check of the recorded optimum in COPASI. Objective of the exported three-condition Peta problem at FungMod’s recorded optimum, after COPASI’s local fit from that point, and at the end of COPASI’s fits from ten random starts (log axis). The recorded outcome is quoted in the manifest.

6 Limitations and claims not made

- One organism, one substrate, one laboratory’s data; no transfer to another strain, substrate or condition has been tested. A survey found no open raw-data deposit of a comparable submerged cellulose culture; figure digitisations with explicit digitisation uncertainty are the only candidates and need full-text review before use.
- Every error model is an assumption because the deposit holds no replicates; the noise multiplier absorbs model misfit and measurement noise together.
- Nutrient, oxygen and maintenance physiology are absent from the organism case; solid-substrate accessibility is a placeholder; there is no spatial mycelium.
- The least-squares optimiser’s finite-difference step was undeclared until the cross-solver check found it stopping 1.3 percent above the minimum; the public calibration API still uses scipy’s default step and is not used by any recorded result in this paper.
- The stage B chains did not reach the declared convergence rule within the plan’s compute cap; their verdicts are provisional.
- Software verification (parity tests, reference simulators, cross-solver agreement) is not empirical validation, and this paper makes no predictive biological claim.

7 Reproducibility and availability

FungMod is MIT-licensed at <https://github.com/felixlaga/FungMod> with documentation at <https://fungmod.readthedocs.io/>. Each study above has a runner script, a frozen plan, a README and checksummed results. Every table and figure in this paper is generated from those result files by one command, `python scripts/reproduce_paper.py tables`, which writes the tables as $\text{\LaTeX}$ fragments that this manuscript includes and the figures as PDF files, together with a manifest naming each table’s and figure’s source files, their SHA-256 digests and the numbers the text quotes; the test suite fails if the committed tables, figures, the manifests or

the sources drift apart, or if the manuscript stops including one of them. The same command has costlier tiers: `check` and `verify` (seconds to a minute) recompute the digest chains, the compiled-core objective at the recorded cross-solver optimum and the stationarity of the recorded baseline fit without re-running any study; `stage-a` re-runs the least-squares stage and the COPASI reproduction (about two hours) and compares them with the recorded summaries; `full` also re-runs the Bayesian study and the posterior chains (a day of compute) and compares verdict-level fields, because seeded chains can differ in individual samples across platforms. The runtime dependency closure is pinned in `requirements-lock.txt`; continuous integration builds the wheel, downloads that closure into a wheelhouse and installs the wheel with network access disabled, then runs a virtual experiment from outside the checkout. A tagged, archived release with a DOI is *pending*.

8 AI assistance disclosure

The code, studies and this draft were produced with substantial assistance from an AI coding assistant (Claude Code, Anthropic) directed by the author, under the repository’s written rules on provenance, maturity labels and forbidden shortcuts, with every scientific study run under a plan frozen before its first fit. Human review of the biological mapping and of this text is outstanding and is not claimed here.

Acknowledgements

The Gelain et al. deposit (CC BY 4.0) made the organism case possible. The COPASI, libSBML, PEtab and SciPy communities provided the tools the cross-solver study depends on.

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